R3.4 Electron-pair sharing reactions

Nucleophilic substitution reaction mechanism (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- (R3.4.9) Describe and explain the mechanisms of the reactions of primary and tertiary halogenoalkanes with nucleophiles.
- (R3.4.9) Explain the stereospecific nature of S_N2.
- (R3.4.10) Predict and explain the relative rates of the substitution reactions for different halogenoalkanes (RCl, RBr, RI).

Section R3.4.1–2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nucleophilic-substitution-reaction-id-43490/) describes halogenoalkanes undergoing substitution reactions in which a halogen atoms is replaced by a nucleophile. This can be illustrated by an orange cat replacing a grey cat that was sleeping in a bed. The grey cat represents a halogen atom in a halogenoalkane compound. The halogenoalkane is a molecule with a carbon bonded to a halogen atom like a chain with one cat at one end. Imagine there is a sleepy grey cat (halide ion) comfortable in her spot, enjoying a peaceful nap. The energetic and playful orange cat (nucleophile) is eager to explore and spots the bed with the grey cat in it. The orange cat is intrigued, sneaks up and quietly approaches the bed out of curiosity. With a sudden burst of energy, the orange cat is kicking out and jumps into the bed, startling the grey cat. The orange cat settles into the bed, and the grey cat is now displaced from the comfortable spot. The orange cat has replaced the grey cat in the bed. The halogen atom is 'kicked out' or replaced by the nucleophile.

Nucleophilic substitution reactions can occur through two possible mechanisms. Imagine that you have two different halogenoalkanes, for example chloromethane (CH_3Cl) and 2-chloro-2-methylpropane ($(CH_3)_3CCl$), reacting with hydroxide ions to produce an alcohol (**Figure 1**). The characteristics of the products are different as the reaction occurs through different substitution reaction mechanisms. What factors might influence the outcome of the reaction for both halogenoalkanes? Is it related to the structure of the halogenoalkanes or the reaction conditions? What are the differences between the two reaction mechanisms?

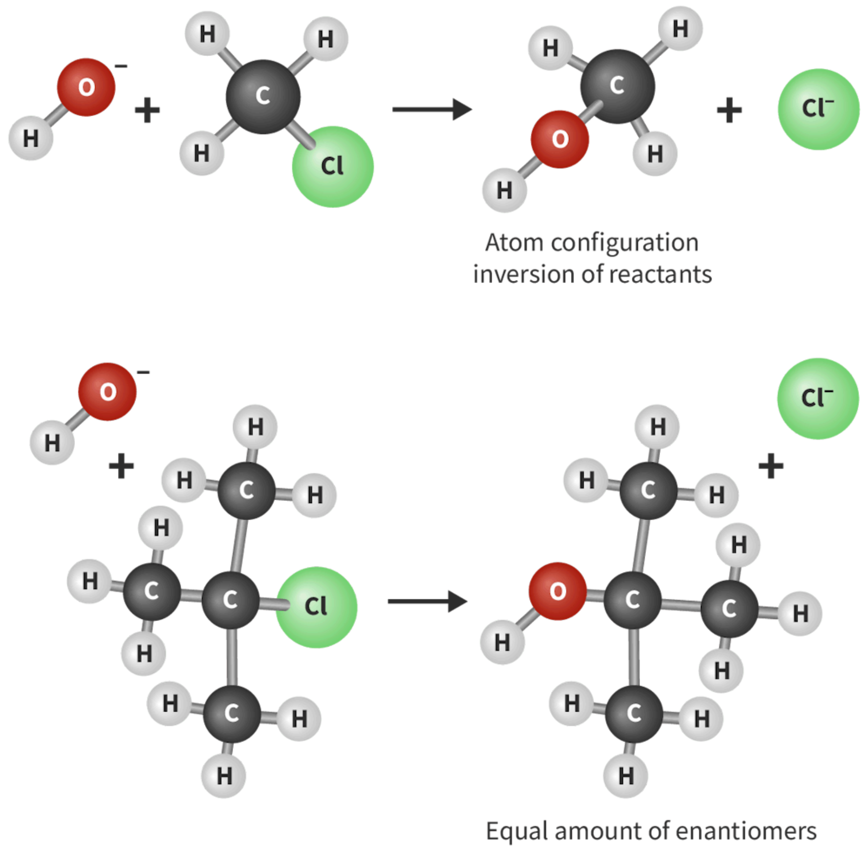

Figure 1. Alcohol formation from different types of halogenoalkanes through different mechanisms.

More information for figure 1

The image shows two nucleophilic substitution reactions of halogenoalkanes with hydroxide ions, illustrating different mechanisms and outcomes.

1. Top Reaction:
2. Reactants: A chloromethane molecule (CH3Cl) and a hydroxide ion (OH^-).
3. Process: Hydroxide ion attacks the carbon atom bonded to chlorine in chloromethane, resulting in inversion of atom configuration.
4. Products: Methanol (CH3OH) and a chloride ion (Cl^-).
5. Note: The reaction results in atom configuration inversion of the reactants.
6. Bottom Reaction:
7. Reactants: A 2-chloro-2-methylpropane molecule ((CH3)3CCl) and a hydroxide ion (OH^-).
8. Process: Hydroxide ion attacks the carbon atom bonded to chlorine, resulting in a different intermediate.
9. Products: Tertiary butyl alcohol ((CH3)3COH) and a chloride ion (Cl^-).
10. Note: The reaction leads to equal amounts of enantiomers due to the formation of a racemic mixture.

[Generated by AI]

The nature of nucleophilic substitution reaction mechanisms depends on the types of halogenoalkane involved in the reaction. Section R3.4.1–2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-

753301/book/nucleophilic-substitution-reaction-id-43490/) introduces the way in which the electron pairs are moved from the nucleophile attacking the partially positively charged carbon atom, causing the halogen atom to leave the group as a leaving group. As a result, the carbon–nucleophile bond and halogen ion are formed. However, the process of this reaction occurs in a series of elementary steps, and two major steps are identified: breaking the C–X bond (departing of the halogen) and forming the C–Nu bond (nucleophilic attack).

The step in which the C–X bond breaks between the carbon atom and the halogen atom results in the departure of the halogen atom, often referred to as the 'leaving group'. The other step is when the nucleophile attacks the partially positive carbon atom as it has a polar bond with the halogen atom. The nucleophile is an electron rich species, attacking the carbon atom by donating a pair of electrons to form a new bond with the carbon atom. Which step happens first? What factors might determine the order of the steps?

The structure of the carbon atom to which the halogen group is bonded (primary, secondary or tertiary) (see section S3.2.6 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structural-isomerism-id-45665/)) plays a significant role in identifying whether the reaction will proceed through different mechanisms, S_N1 or S_N2 reaction mechanism. S_N means substitution nucleophilic: the number 1 means unimolecular and number 2 means bimolecular. How does the structure of the halogenoalkanes affect the reaction mechanism? The two reaction mechanisms differ in their steps in which the leaving group departs. The nature of the reaction mechanisms depends on several factors such as the type of halogenoalkane involved and the structure of the carbon–halogen bond.

S_N2 (nucleophilic bimolecular substitution)

This reaction mechanism occurs by the breaking of the carbon–halogen bond due to nucleophilic attack and the simultaneous departure of the leaving group. This means that the nucleophilic substitution proceeds in one step and occurs in primary halogenoalkanes. Primary halogenoalkanes have a structure in which the carbon–halogen bond is attached to only one other carbon atom. This results in less steric hindrance as hydrogen atoms are relatively small. The space around the carbon–halogen bond is unobstructed as it has only one carbon atom attached, making it easily accessible to the nucleophile.

Study skills

The presence of three alkyl groups in the tertiary halogenoalkanes means a nucleophile may bump into an electrophile rather than react, and is called steric hindrance. Therefore, the carbon—halogen bond is broken in the first step forming the carbocation intermediate to proceed with the reaction. Below is an interactive artwork that compares the steric hindrance experienced by primary, secondary and tertiary halogenoalkanes when undergoing nucleophilic attacks. Click on the arrows in **Interactive 1** to view the different stages.

Interactive 1. Steric Hindrance Hinders the Nucleophilic Attacks in Primary, Secondary and Tertiary Halogenoalkanes.

More information for interactive 1

The interactive illustrates the effect of steric hindrance on nucleophilic substitution in primary, secondary, and tertiary halogenoalkanes in four sections. The common text on the left of each section reads as follows. Steric hindrance hinders the nucleophilic attacks in primary, secondary, and tertiary halogenoalkanes. Section 1. At the top, molecular structures of the halogenoalkanes depict the increasing steric hindrance as the number of alkyl groups around the carbon-halogen bond increases. The bottom section uses space-filling models to emphasize how alkyl groups obstruct nucleophilic attack. In primary halogenoalkanes, nucleophiles can attack with relative ease, whereas secondary halogenoalkanes experience some restriction due to bulkier alkyl groups. In tertiary halogenoalkanes, steric hindrance is most pronounced, effectively preventing nucleophilic attack.
Section 2. Primary halogenalkane. The diagram focuses on the steric hindrance in primary halogenoalkanes during nucleophilic substitution. The left side presents a ball-and-stick molecular structure, showing the halogenoalkane with a nucleophile (Nu) approaching the carbon bonded to a chlorine atom. Steric hindrance from surrounding hydrogen atoms is indicated. On the right, a space-filling model visually emphasizes how the alkyl group partially obstructs the nucleophilic attack. Despite some hindrances, primary halogenoalkanes are still relatively accessible to nucleophiles compared to their secondary and tertiary counterparts.
Section 3. This diagram illustrates the steric hindrance in secondary halogenoalkanes during nucleophilic substitution. On the left, a ball-and-stick molecular structure depicts a secondary halogenoalkane with a nucleophile (Nu) approaching the carbon bonded to a chlorine atom. The presence of two alkyl groups increases steric hindrance, restricting nucleophilic attack. On the right, a space-filling model visually highlights how the bulky alkyl groups further obstruct the nucleophile's access to the reactive carbon, making substitution more challenging compared to primary halogenoalkanes.
Section 4. This diagram illustrates steric hindrance in tertiary halogenoalkanes during nucleophilic substitution. On the left, a ball-and-stick molecular structure represents a tertiary halogenoalkane, where a nucleophile (Nu) attempts to approach the central carbon bonded to a chlorine atom. The three surrounding alkyl groups create significant steric hindrance, preventing nucleophilic attack. On the right, a space-filling model further emphasizes how the bulky alkyl groups completely block access to the reactive carbon, making direct nucleophilic substitution highly unlikely.

For example, in the reaction between chloroethane and aqueous sodium hydroxide:

$$CH_3CH_2Cl(aq) + OH^-(aq) \rightarrow CH_3CH_2OH\ (aq) + Cl^-\ (aq)$$

the mechanism of this reaction is a one-step concerted reaction in which the hydroxide ion as the nucleophile is attracted to a slightly positive carbon and forms an unstable transition state (**Interactive 1**). The electronegativity of the leaving group plays a significant role in this process. As the leaving group is more electronegative than carbon, it has a greater ability to attract a pair of electrons towards itself, causing the leaving group to depart leaving the carbon atom to be electron-deficient and thus more positively charged. Also, a one-step concerted reaction refers to a type of reaction mechanism, which means a chemical reaction in which all bond breaking and bond making occur in a single step without any intermediate stages or unstable products. All the atoms involved in the reaction are rearranged and their positions generate the products at the same time.

The chemical structure illustrated in the square bracket in **Interactive 2** is not considered an intermediate product. It continues with the departure of chloride atoms as a chloride ion and produces alcohol. Click on the buttons to view the mechanism stages.

Interactive 2. The S_N2 Reaction Mechanism.

More information for interactive 2

The interactive displays The S_N2 reaction mechanism through seven slides.
Slide 1: A carbon atom is bonded to two hydrogen atoms in wedge and dash bonds. The carbon atom is also bonded to one methyl group (CH_3) and a chlorine (Cl) atom. The carbon atom has a positive charge and it is mentioned using δ+ and the chlorine atom has a negative charge and is represented as δ-. The bond between central carbon and chlorine is pointed using an arrow as a polar bond. The electrons in the polar bond are mentioned as two dots, one from carbon and one from chlorine.
Slide 2: The same structure is redrawn. An arrow is drawn to the carbon atom with positive charge, indicating that it is an Electron-deficient carbon atom.
Slide 3: A hydroxy group (OH) with a lone pair of electrons attacks the electron-deficient carbon atom from the left side.
Slide 4: The Transition state of the reaction is drawn. The hydroxy group is now bonded to the central carbon along with two hydrogen atoms, a methyl group and chlorine. The bonds of the hydroxy group and chlorine atom to the central carbon are in dashed lines. The whole structure is drawn inside a bracket with a negative charge on top of the bracket.

Slide 5: The chlorine atom leaves from the transition state. This is indicated by an arrow drawn from the bond towards the chlorine atom.
Slide 6: The hydroxy group, OH, is now bonded to the central carbon atom, and the chlorine ion is given as the product with four pairs or lone pairs and a negative charge.
Slide 7: The overall chemical reaction. The first structure on the left shows a hydroxide ion, OH-, attacking a carbon bonded to a chlorine atom, the carbon atom has δ+ charge, and the chlorine atom has δ- charge. A curved arrow indicates bond breaking between the carbon and chlorine. The middle structure represents the transition state, where the hydroxide and chlorine are both partially bonded to the carbon. The right structure shows the final products, where the hydroxide is fully bonded to the carbon, forming an alcohol, and the chloride ion, Cl-, is released. The first step is labelled slow, and the second step is labelled fast.

The S_N2 reaction mechanism is often called a 'backside attack' as the reaction occurs because the nucleophile attacks the partially positively charged carbon atom that is located on the opposite side of the halogen atom. The nucleophile has a lone pair of electrons, making it electron rich and the leaving group is typically an atom or group with a high electronegativity. What kind of interactions would you expect between them as they are both electron-rich species and negatively charged?

They both will experience electrostatic repulsion between them. This causes the nucleophile to be repelled by the electron-rich leaving group if it approaches from the same side. The nucleophile approaches the positively charged carbon atom from the side opposite to the leaving group to minimise repulsion.

Once the nucleophile attacks the partial positively charged carbon atom, it displaces the leaving group and simultaneously forms a new bond. This process can be visualised as a 'bumping' effect similar to Newton's cradle, as the nucleophile pushes the leaving group out of the molecule. The incoming nucleophile represents the striking ball and the leaving group acts as the ball on the end. When the nucleophile approaches the partial positively charged carbon atom, the momentum is gained and transferred to the leaving group, causing it to be 'bumped' out of the molecule.

Furthermore, the backside attack by the nucleophile causes the inversion of the configuration in which the other atoms around carbon are being arranged, because the nucleophile needs to pass through the centre of the existing groups surrounding the carbon where both the nucleophile and the leaving group are partially bonded to the carbon. At this stage, the carbon atom configuration is not yet lost as the bond is not fully formed or broken. The new bond is formed when the leaving group completely leaves. The S_N2 reaction mechanism is said to be stereospecific due to the three-dimensional arrangement of reactants being converted into the three-dimensional stereoisomeric products. The inversion of the configuration can be visualised as an umbrella that is being blown inside-out (**Figure 2**).

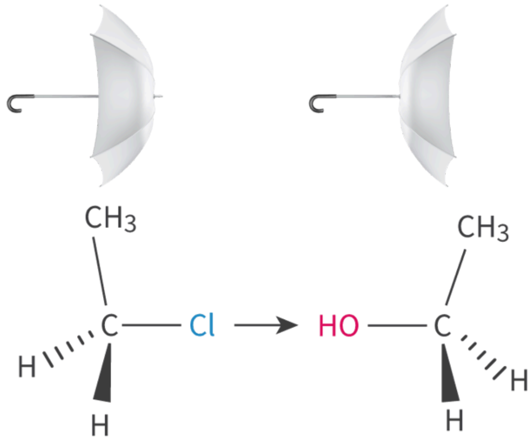

Figure 2. The inversion of atoms configuration around the atom carbon during the S_N2 reaction.

More information for figure 2

The image is a diagram illustrating the SN2 reaction mechanism. It features two configurations of a methyl chloride (CH3Cl) molecule transforming into a methyl alcohol (CH3OH) molecule, symbolizing the inversion process around the central carbon atom. The drawing includes an umbrella on each side to metaphorically represent this inversion, which appears as if an umbrella is being turned inside-out. On the left, a chlorine atom (Cl) is bonded to the carbon, indicated by blue text, with three hydrogen atoms (H) and one methyl group (CH3) connected to it in a tetrahedral arrangement. An arrow points to the right, where a hydroxide group (OH), indicated by red text, takes the place of the chlorine atom, maintaining the tetrahedral geometry but inverted. This represents the nucleophilic substitution that characterizes the SN2 reaction.

[Generated by AI]

The S_N2 reaction mechanism is a single-step reaction in which the rate of reaction is significantly influenced by the concentration of both the halogenoalkane and the nucleophile. It means that if the concentration of either reactant is doubled, the rate of reaction will also double. This is the important characteristic of the S_N2 reaction mechanism, where both reactants are essential for the reaction to occur. The S_N2 reaction is a bimolecular reaction where the rate of reaction is dependent on the collision between two molecules (nucleophile and halogenoalkane). Thus, it can be described as a first-order reaction with respect to both the halogenoalkane and the nucleophile (second-order overall) (section R2.2.6 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/reaction-mechanisms-hl-id-45509/)). The reaction rate between halogenoalkane and nucleophile is expressed as:

$$\text{Rate} = k[\text{halogenoalkane}][\text{nucleophile}]$$

The energy profile for the reaction between chloroethane and the hydroxide ion is illustrated in **Figure 3**. A weak transition state is formed in which a weakened carbon–chlorine bond is broken simultaneously with the formation of a new bond between carbon and oxygen atoms that requires a high activation energy.

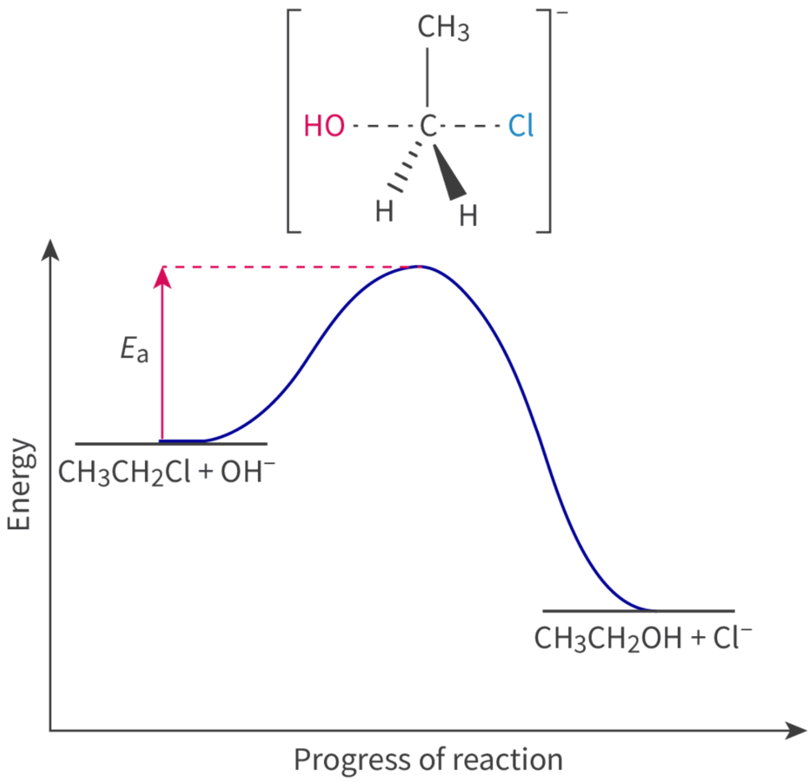

Figure 3. S_N2 reaction energy profile.

More information for figure 3

The image is a graph representing the energy profile of an SN2 reaction involving chloroethane (CH3CH2Cl) and hydroxide ion (OH−). The X-axis is labeled 'Progress of reaction,' while the Y-axis is labeled 'Energy.' The graph depicts a curve with an initially low energy level where CH3CH2Cl and OH− reactants are present. This is followed by a peak representing the transition state, where the activation energy (Ea) is measured. At this point, a temporary complex is formed with partial bonds involving carbon (C), hydroxide (OH), and chlorine (Cl) atoms. The high energy indicates the energy required to break the carbon-chlorine bond while forming a new bond between carbon and oxygen. Once the transition state is surpassed, the energy decreases to a lower level, where the products CH3CH2OH (ethanol) and Cl− are formed. This illustrates the stability of the products compared to the initial reactants.

[Generated by AI]

S_N1 (nucleophilic unimolecular substitution)

This is different from the previous nucleophilic substitution reaction mechanism, as the S_N1 reaction involves two steps and occurs in tertiary halogenoalkanes. Why is the S_N1 reaction more likely to occur? Would there be any room for the nucleophile to attack in the case of tertiary halogenoalkanes?

In the case of tertiary halogenoalkanes, the nucleophile can only attack the halogenoalkanes if the leaving group has left the molecule. It is started by dissociating the leaving group from the molecule to create room and a position for the nucleophile to occupy.

The first step that occurs is the formation of a carbocation: a carbon with a positive charge through the breaking of its carbon–halogen bond in which an electron pair between these two atoms is taken by the halogen group and then forms a carbocation intermediate and halide ion. For example, consider a reaction between 2-chloro-2-methylpropane and aqueous sodium hydroxide. The first step of this reaction results in the formation of a carbocation intermediate with a temporary positive charge and chloride ion as illustrated in **Interactive 3**. Click on the button to view the different stages.

Interactive 3. The First Step of the S_N1 Reaction Mechanism.

More information for interactive 3

An interactive displays the first step of the S_N1 reaction mechanism in three steps in three slides.

Slide 1: It illustrates the tertiary carbon atom is bonded to three methyl groups (CH_3) and a chlorine atom (Cl). The carbon is marked as partially positive (δ^+) and the chlorine is partially negative (δ^-). A curved arrow points from the bond between carbon and chlorine to the chlorine atom, indicating the departure of the chloride ion (Cl^-), forming a carbocation intermediate.

Slide 2: In this step, a tertiary carbocation is formed where a central carbon, bonded to three methyl groups (CH_3), carries a positive charge (C^+). A chloride ion (Cl^-) with a full octet of electrons is shown as the leaving group.

Slide 3: In this step, a tertiary alkyl chloride undergoes ionization: the central carbon, bonded to three methyl groups and a chlorine atom, shows partial charges ($C\ \delta^+$ and $Cl\ \delta^-$). A curved arrow points from the C–Cl bond to the chlorine, indicating heterolytic cleavage. The reaction is labeled as 'slow,' forming a carbocation intermediate (Cl^+) and a chloride ion (Cl^-).

The second step of this reaction mechanism involves a nucleophilic attack on a carbocation intermediate resulting in a new bond formation. The hydroxide ion donates a pair of electrons to the electron-deficient carbon atom, leading to the formation of 2-methyl-propan-2-ol, which is a tertiary alcohol (see section S3.2.6 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structural-isomerism-id-45665/)) as shown in **Interactive 4** next. Click on the button to view the stages.

Interactive 4. The Second Step of the S_N1 Reaction Mechanism.

More information for interactive 4

An interactive displays the second step of the S_N1 reaction mechanism in three steps in three slides.

Slide 1: In this interactive, a hydroxide ion (OH^-) donates a lone pair to a tertiary carbocation, forming a new C–O bond. The diagram shows the curved arrow representing a nucleophilic attack on the positively charged carbon center.

Slide 2: Product formed in the second step of the S_N1 reaction mechanism. A tertiary carbon atom is bonded to three methyl groups and one hydroxyl group (OH), indicating the nucleophile (OH^-) has successfully bonded to the carbocation to form the final alcohol product.

Slide 3: The interactive showing a hydroxide ion (OH^-) attacking a tertiary carbocation through nucleophilic attack. A curved arrow points from the lone pair on the oxygen to the positively charged carbon. The fast reaction leads to the formation of a tertiary alcohol product.

The three alkyl groups attached to the carbon atom stabilise the carbocation in the tertiary halogenoalkanes. In the case of carbon–carbon bonds, the similar electronegativities cause the bonds to be non-polar. As a result, the positive charge of the carbocation due to the electron pair removal can be shared across all three carbon–carbon bonds in the molecule. Therefore, tertiary halogenoalkanes are more stable compared to secondary halogenoalkanes which are more stable than primary halogenoalkanes. This stabilising effect perseveres for long enough time enabling the carbocation to proceed to the second step of the reaction.

Nature of Science

Aspect: Theory

Mechanistic models such as S_N1 and S_N2 are used to describe chemical reactions that involve nucleophilic substitution. Their mechanisms are widely used in the pharmaceutical and food industry. The choice between the S_N1 and S_N2 mechanisms depends on the aim of proceeding with the reaction. The model of these reaction mechanisms helps predict the characteristics of species involved and other factors that influence the outcome of the reaction, such as the nature of species involved, the solvent and so on, on the rate of nucleophilic substitution. The theories about these two mechanisms only can be predicted and developed by collecting data including information about both reactant and product stereochemistry.

The concentration of which reactant determines the rate of the S_N1 reaction mechanism? Think about your answer before revealing the solution.

The rate of the S_N1 reaction mechanism is determined by the concentration of the halogenoalkane because the nucleophile does not participate in the rate-determining (slowest) step (see section R2.2.6 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/reaction-mechanisms-hl-id-45509/)).

The reaction rate is expressed here.

Rate = k[halogenoalkane]

The energy profile for this reaction is illustrated in **Figure 4**. The activation energy in the first step is relatively high compared to the activation energy in the second step. This occurs due to the initial stage involving an endothermic process in which the bonds are broken, while in the second step there is a strong attraction between the oppositely charged ions (carbocation and halide ion).

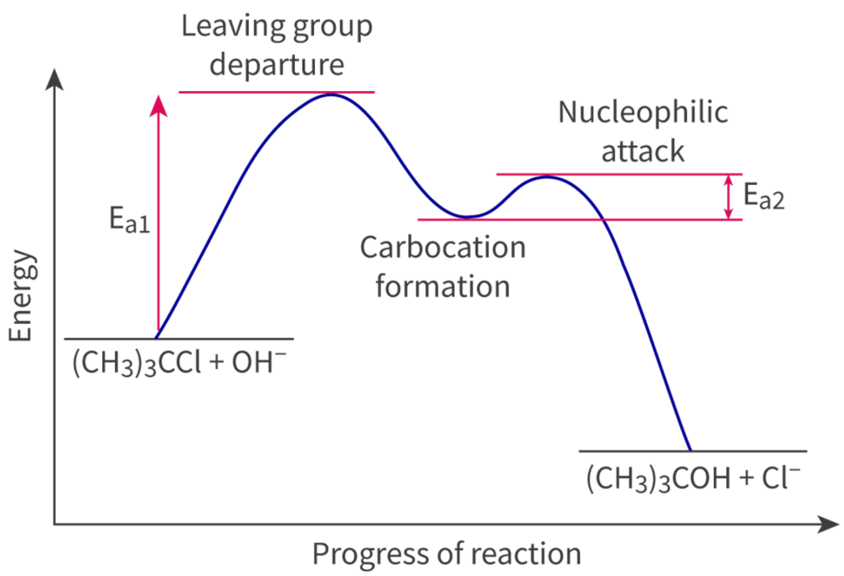

Figure 4. The second step of the S_N1 reaction mechanism.

More information for figure 4

This is a graph representing the energy profile of an SN1 reaction mechanism. The X-axis is labeled as "Progress of reaction" and the Y-axis is labeled as "Energy." The graph starts with the reactants (CH3)3CCl + OH−. As the reaction progresses, there is an initial rise in energy marked as the leaving group departure, labeled with an activation energy Ea1. This is an endothermic process where bonds are broken. Following this, there is a decrease in energy which corresponds to the carbocation formation. The energy reaches a peak again, indicating the nucleophilic attack, which has a smaller activation energy Ea2 compared to Ea1. Finally, the product of the reaction is (CH3)3COH + Cl−.

[Generated by AI]

Because the carbocation intermediate that is formed in the S_N1 reaction does not show stereospecificity, it allows the nucleophile to approach from any side with equal probability. This results in the formation of a mixture of optical isomers known as a racemic mixture. In section S3.2.7b (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/stereoisomers-optical-isomerism-hl-id-45667/), a racemic mixture described as being optically inactive and having a 1:1 enantiomer mixture.

Study skills

The reaction mechanism of secondary halogenoalkanes cannot be explained in the two reaction mechanisms. This type of halogenoalkane commonly undergoes a mixture of both mechanisms (S_N1 and S_N2). The preference for a secondary halogenoalkane mechanism such as S_N2 or S_N1 depends on several factors such as the nature of nucleophile and steric hindrance. For example, if the nature of nucleophilic is weak or non-basic, the S_N1 is favoured as the nucleophile is not efficient enough to displace the leaving group directly. A strong nucleophile is required as it will more readily donate its electron pair and overcome the steric hindrance to replace the leaving group in a single step. Also, the steric hindrance of the secondary halogenoalkanes can hinder the direct nucleophile attack making the S_N1 more favourable.

Exercise

Click a question to answer

1. 2-bromo-2-methylpropane will form a tertiary ______ when it reacts with aqueous sodium hydroxide as it is a more stable intermediate. Alcohol is then formed as a result of further attack by the nucleophilic on the intermediate.

Check answers

Higher level (HL)

The identity of halogens as a leaving group

The reaction rate of both S_N1 and S_N2 reactions can increase if it involves a good leaving group. The rate of reaction will be relatively high if the rate-determining step occurs quickly and this is determined by the bond breaking of the carbon–halogen bond heterolytically. The weaker the carbon–halogen bond, the quicker bond breaking will be in the higher rate of reaction.

The fluorine atom that is attached to the carbon of alkane is difficult to react with because the bond between these two atoms is very strong. As you move down the group, the size of halogen atoms increases resulting in a weaker and longer carbon–halogen bond. Therefore, the halogen atom that is easier to dissipate the negative charge is iodine as it is bigger and causes weaker carbon–halogen bonds. Additionally, the polarity of the bond in iodoalkanes is the least polar due to both carbon and iodine having the same electronegativity values.

In **Table 1** is the carbon–halogen bond enthalpy data provided in section 12 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/) of the DP chemistry data booklet.

Table 1. Enthalpy data for carbon halogen bonds.

Bond	Bond energy (kJ mol^{-1})
C—F	492
C—Cl	324
C—Br	285
C—I	228

The order at which a carbon–halogen bond ceases to break (most reactive to least reactive) is:

$$R{-}I > R{-}Br > R{-}Cl > R{-}F$$

In **Interactive 5**, drag and drop the rate determining step, mechanism, order of reaction, reaction rate, carbocation stability, effects of nucleophile and stereochemistry into the correct locations in the table.

	SN1	SN2
Rate determining step		
Mechanism		
Order of reaction		
Reaction rate		
Carbocation stability		
Effects of nucleophile		
Stereochemistry		

Second-order

Rate = k[halogenoalkane][nucleophile]

R–X + Nu^- → R–Nu + X^-

Rate = k[halogenoalkane]

Tertiary

One step

Inversion

Two step

No effect

First-order

Faster rate with stronger nucleophiles

Racemic mixture

R–X → R^+ → R–Nu

Primary

Check

Interactive 5. A comparison of S_N1 and S_N2 reactions.

Theory of Knowledge

Reaction mechanisms are supported with experimental evidence, but cannot yet be proved directly. To what extent does intuition play a role in the proposal of the steps for a mechanism? Can there be scientific truth without direct evidence?

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:**
  - Thinking skills — Designing procedures and models
  - Communication skills — Using terminology, symbols and communication conventions consistently and correctly
- **Time required to complete activity:** 30—45 minutes

- **Activity type:** Pair activity

Your task

We will revisit the cat substitution analogy introduced at the beginning of this section. In pairs, create one analogy for each of the S_N1 and S_N2 reaction mechanisms, discuss factors affecting mechanisms and relate them to a cat substitution scenario. You can illustrate both analogies on a large piece of paper to visually represent them.

Materials

- A large piece of paper
- Markers or coloured pens/pencils.

Some factors you may include:

- Size/age and stability of the cat drawn to represent the leaving group.
- Aggressiveness of the attacking cat drawn to represent the strength of the nucleophile.
- Availability of space for the attacking cat to represent the steric hindrance.
- Persistence of the cat in representing the reaction rate (faster or slower).
- Resistance to the disturbance representing the stability of intermediates formed.

You can include any other factors in as much detail as possible.

Provide examples of reactions based on the analogies you have created and discuss them with other student pairs. Analyse whether they follow both S_N1 and S_N2 mechanisms based on the factors discussed. Summarise the key ideas addressed in the activity, and emphasise the analogies, factors and the relationship to the reaction example.

Other analogies you may want to illustrate:

- Team substitution: the captain of a sports team steps down from the team (players) and a new player wants to join them by substituting one of the existing players.
- The bus ride: a person in a crowded bus gets off at a stop. A new passenger waiting at the bus stop gets on the bus and notices a priority seat for people who need them. The new passenger taps the person sitting there to take the seat.
- You have four textbooks in a backpack/totebag and you realise that you accidentally took your classmate's book by mistake. You prepare to switch books — you remove their book from your bag, then place your book inside OR you place your book inside then remove their book.